EXPERIMENT 9 – Threading

Aim: To create and execute multiple threads using POSIX Threads (Pthreads).

C Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

void *thread_function(void *arg)
{
    int i;

    for(i = 1; i <= 5; i++)
    {
        printf("Thread Executing : %d\n", i);
        sleep(1);
    }

    pthread_exit(NULL);
}

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread_function, NULL);
    pthread_create(&t2, NULL, thread_function, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("All Threads Completed\n");

    return 0;
}
```

Compilation: `gcc threading.c -o threading -pthread`

Output:

```
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
```

Shell Script (Background process simulation):

```
#!/bin/bash

task1()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 1 : $i"
        sleep 1
    done
}

task2()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 2 : $i"
        sleep 1
    done
}

task1 &
task2 &
wait
echo "All Threads Completed"
```

Result: **Threading using POSIX Pthreads was implemented successfully.**
